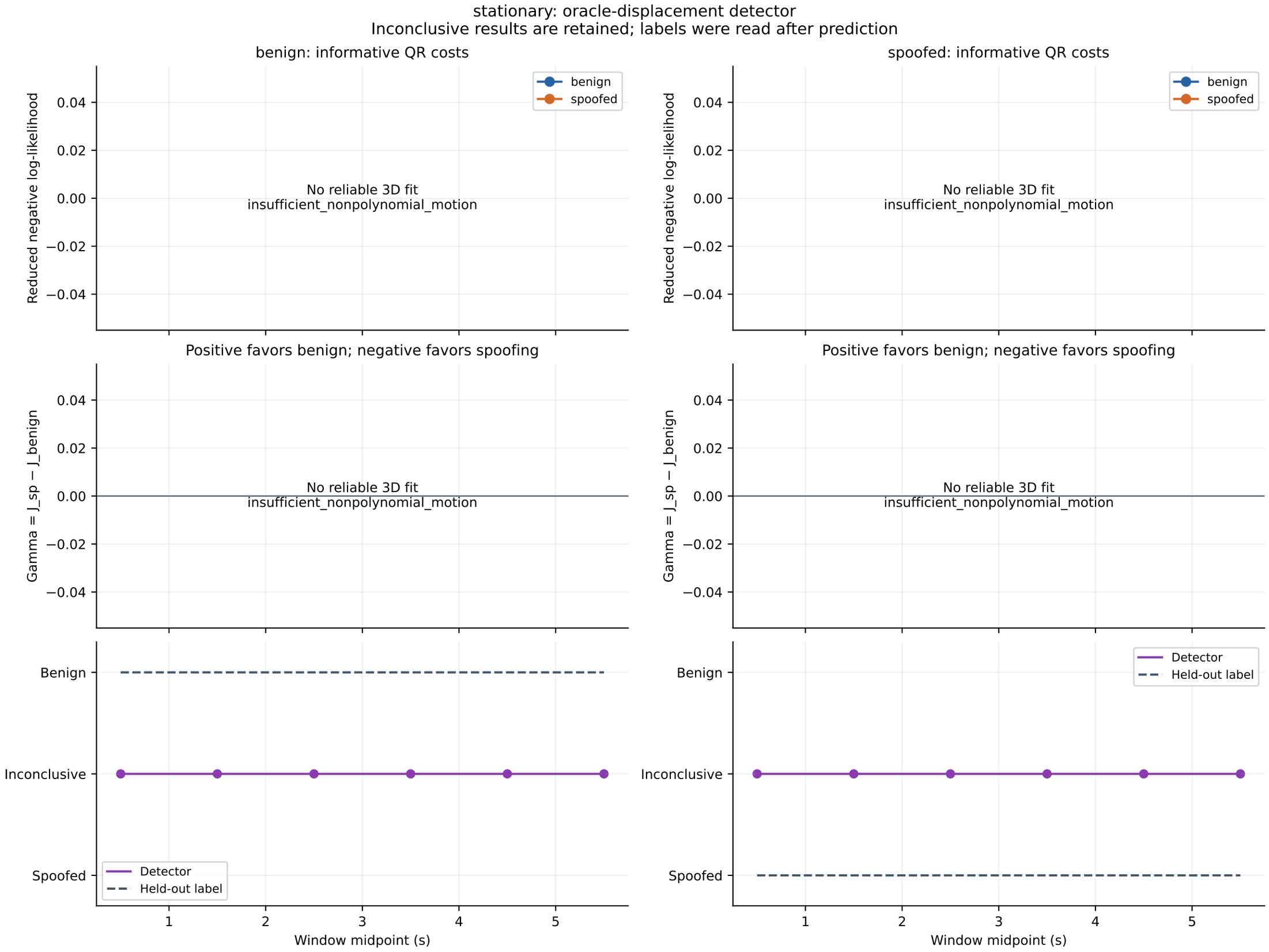
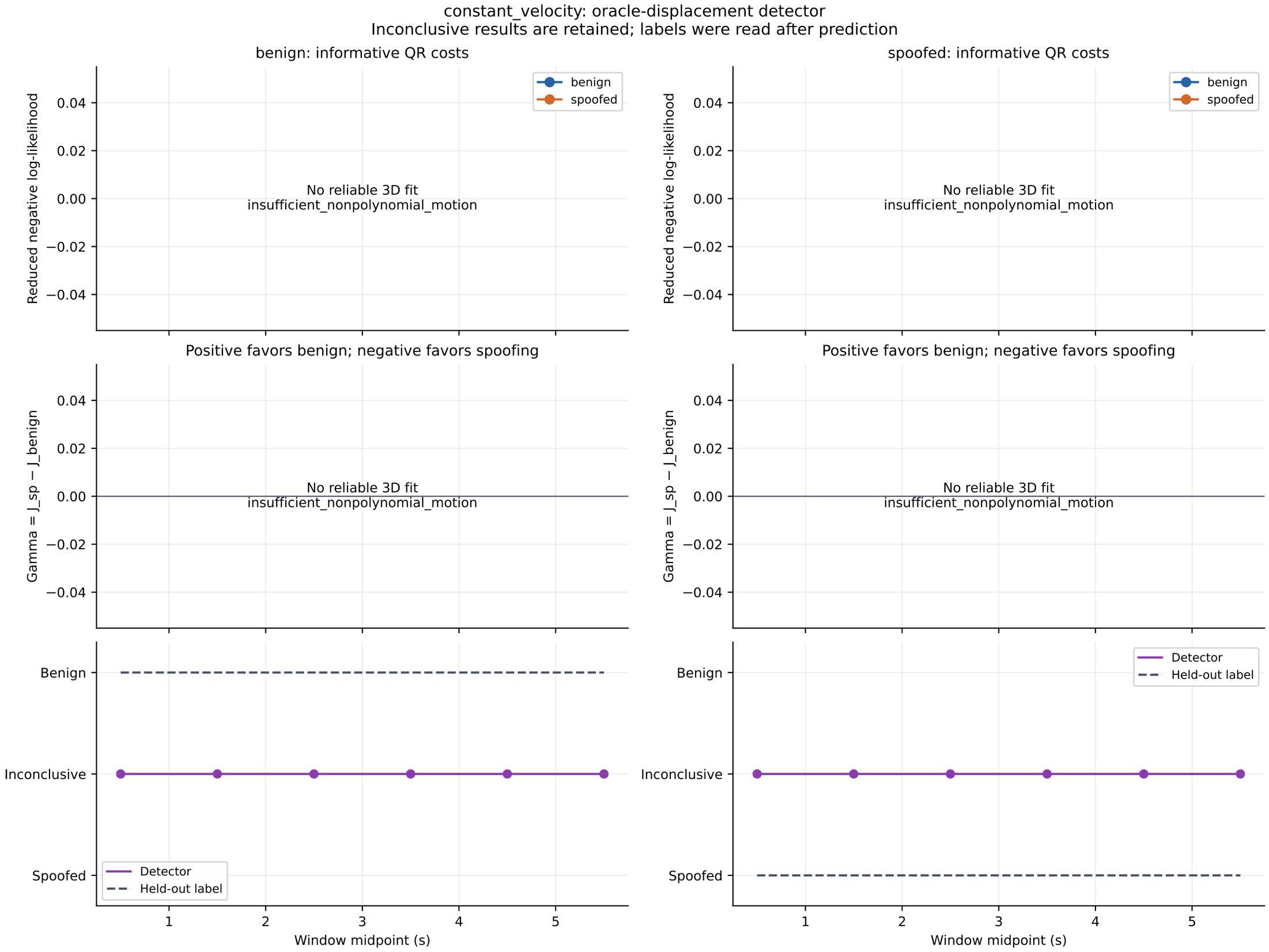
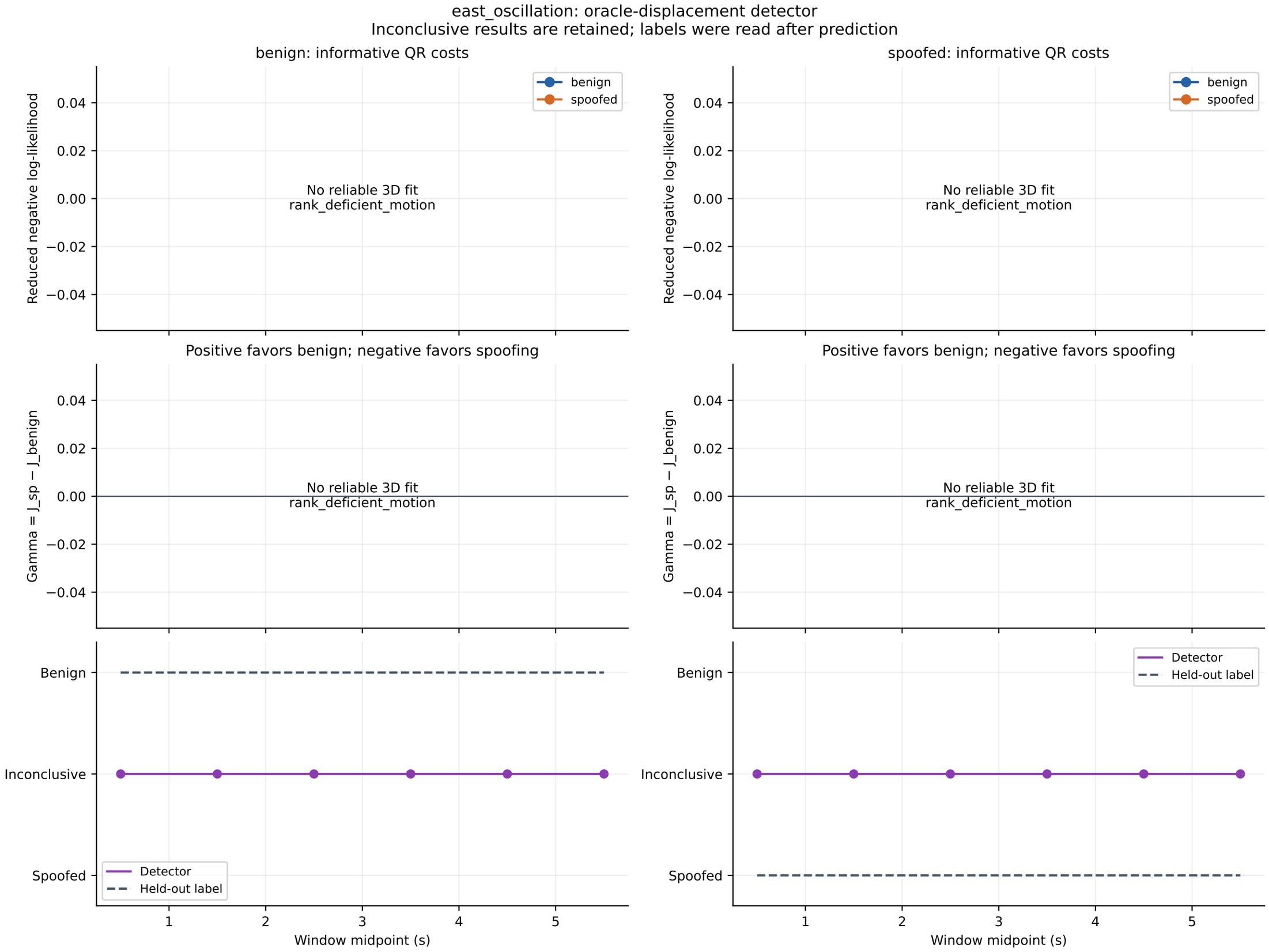
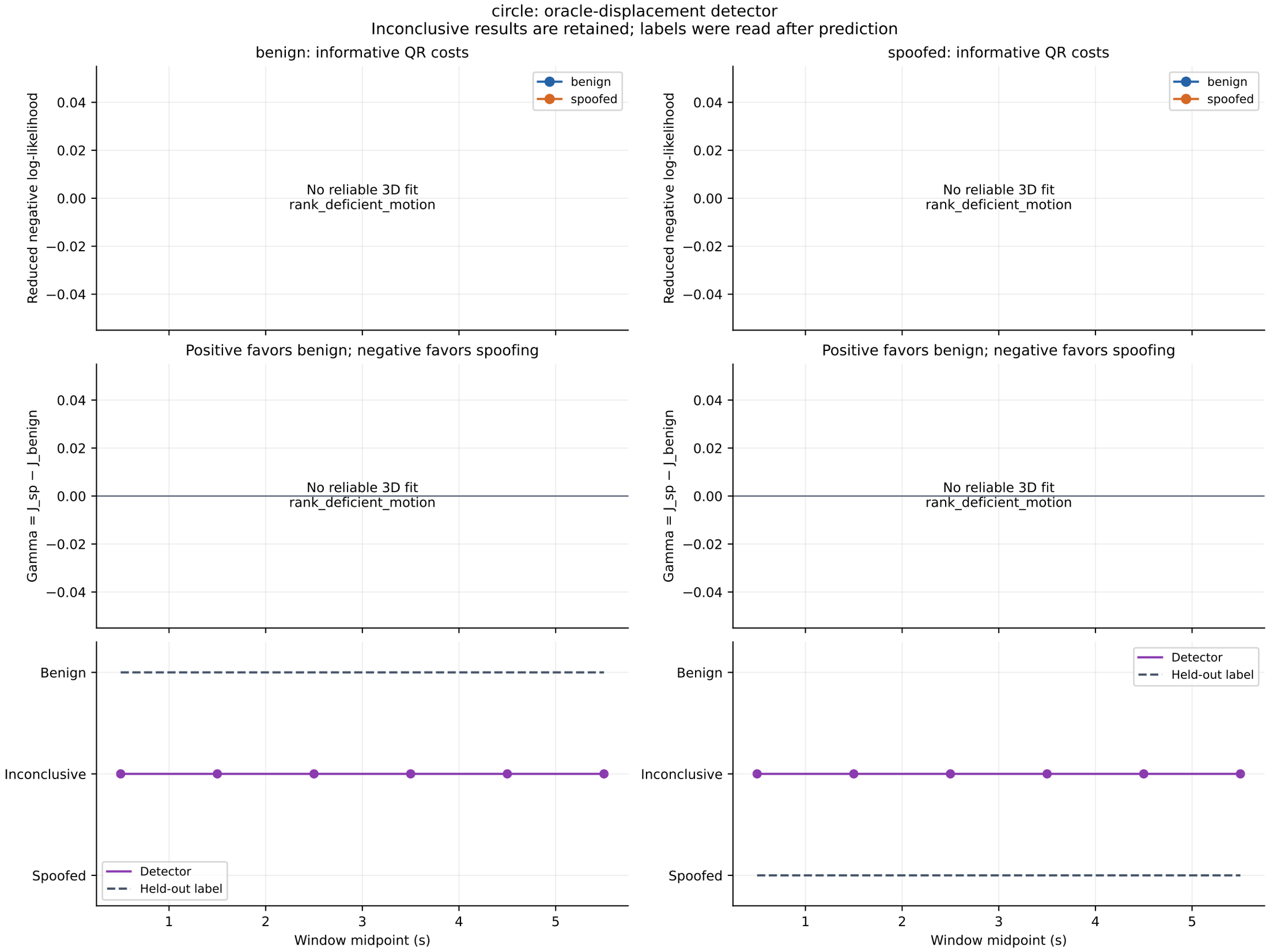
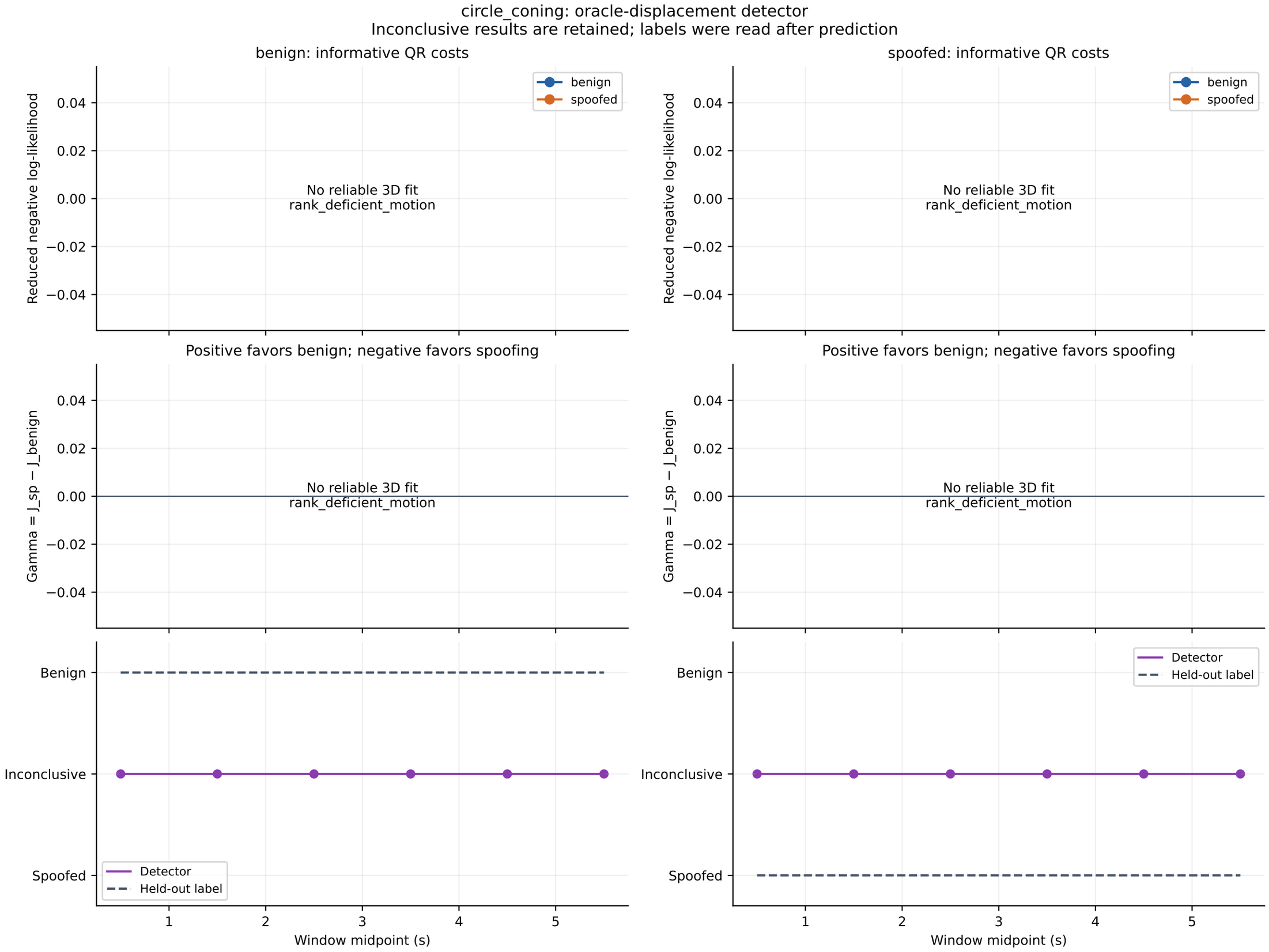
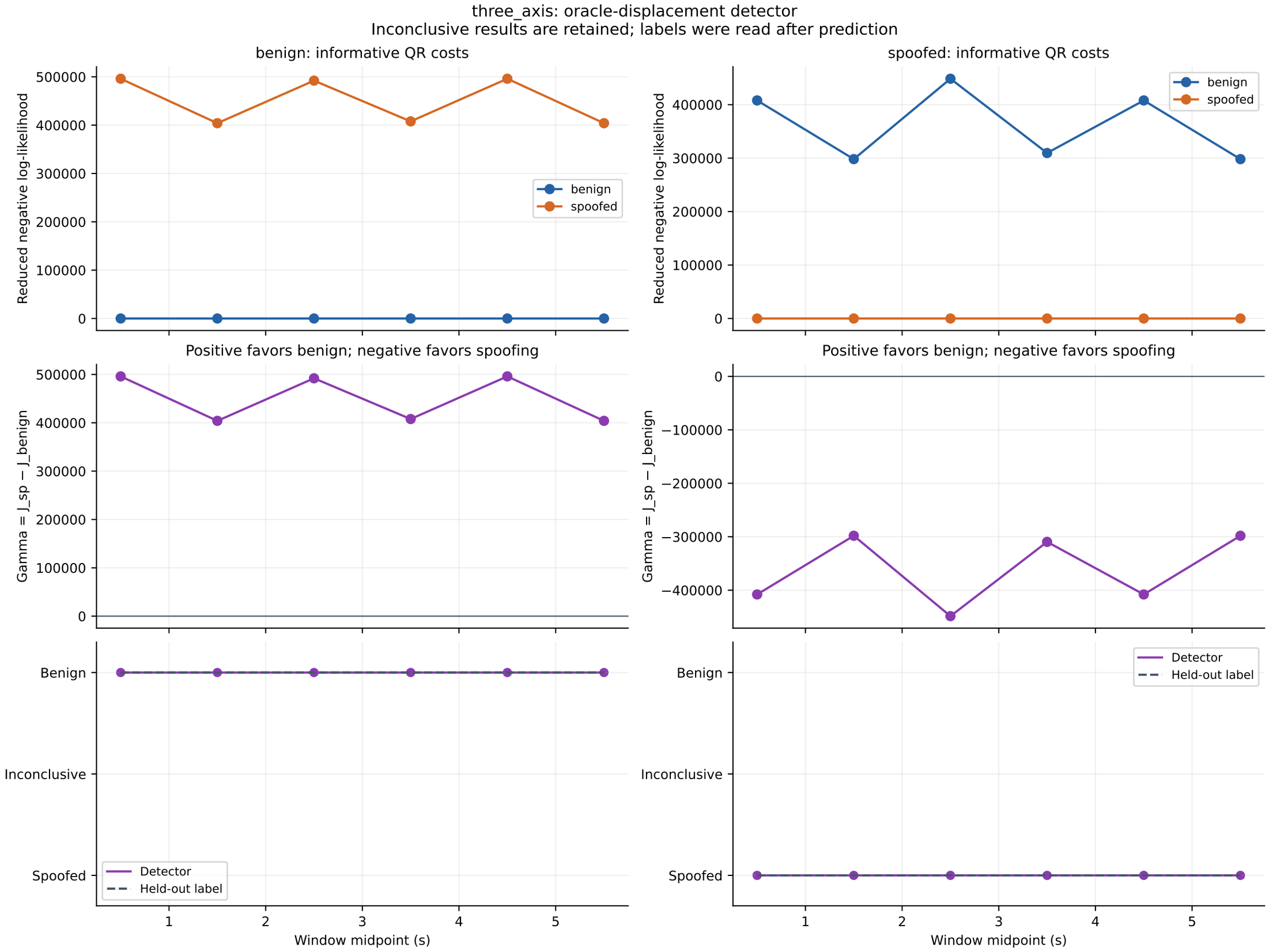

stationary: oracle-displacement detector
Inconclusive results are retained; labels were read after prediction
benign: informative QR costs
spoofed: informative QR costs
Reduced negative log-likelihood
benign
spoofed
No reliable 3D fit
insufficient_nonpolynomial_motion
Positive favors benign; negative favors spoofing
Gamma = J_sp − J_benign
Benign
Inconclusive
Spoofed
Detector
Held-out label
Window midpoint (s)

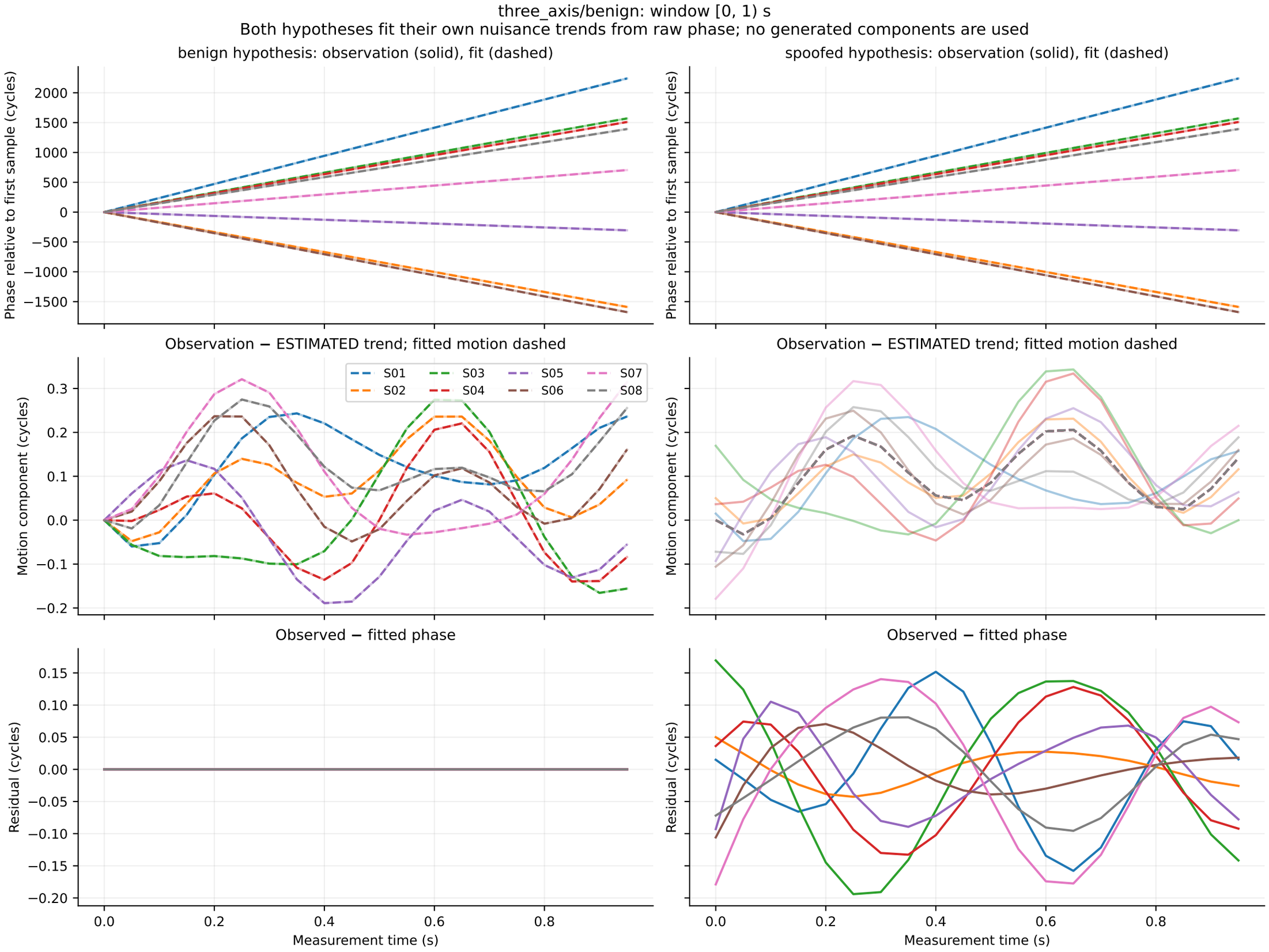

constant_velocity: oracle-displacement detector
Inconclusive results are retained; labels were read after prediction
benign: informative QR costs
spoofed: informative QR costs
Reduced negative log-likelihood
benign
spoofed
No reliable 3D fit
insufficient_nonpolynomial_motion
Positive favors benign; negative favors spoofing
Gamma = J_sp − J_benign
Benign
Inconclusive
Spoofed
Detector
Held-out label
Window midpoint (s)

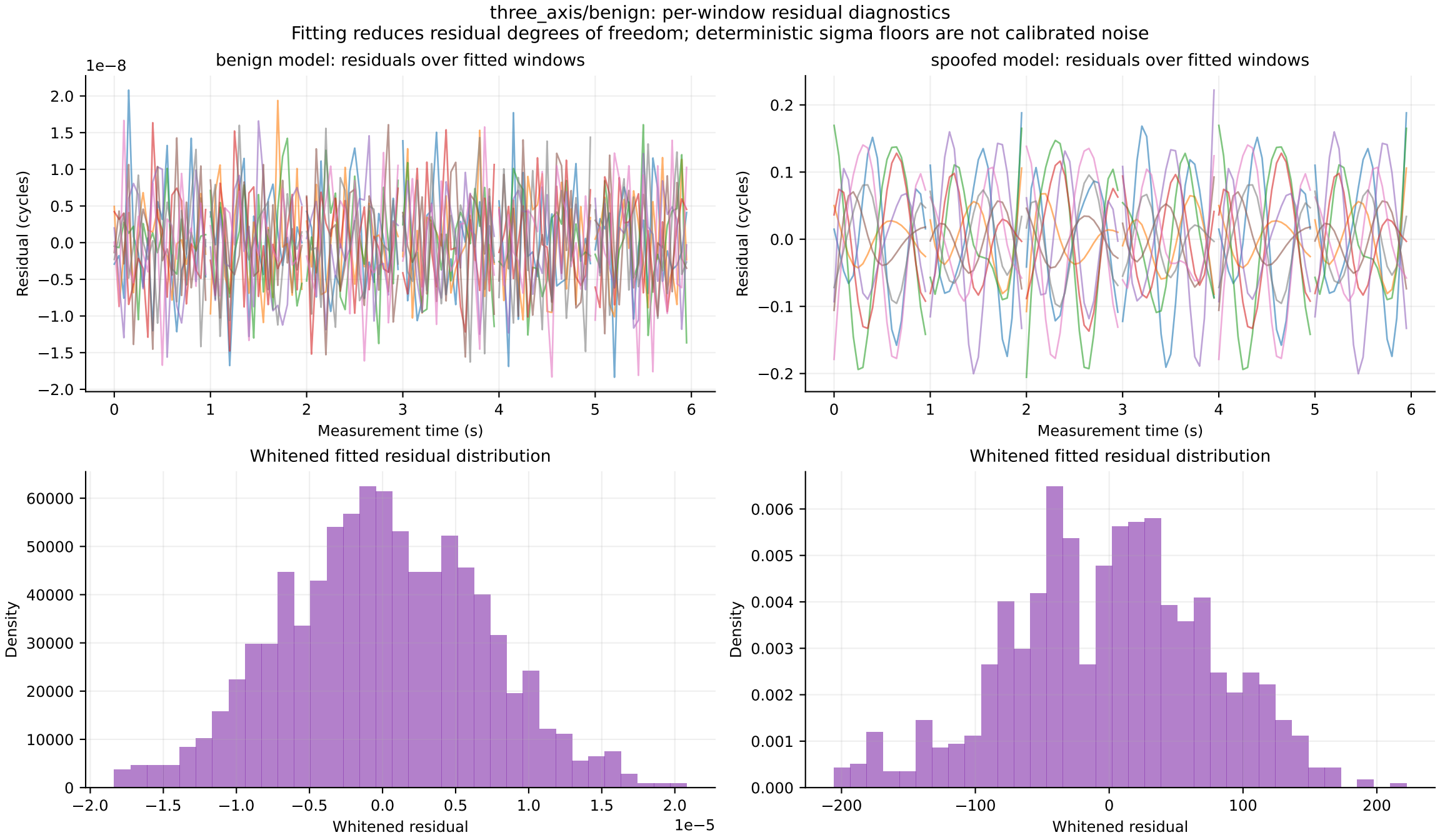

east_oscillation: oracle-displacement detector
Inconclusive results are retained; labels were read after prediction
benign: informative QR costs
spoofed: informative QR costs
Reduced negative log-likelihood
benign
spoofed
No reliable 3D fit
rank_deficient_motion
Positive favors benign; negative favors spoofing
Gamma = J_sp − J_benign
Benign
Inconclusive
Spoofed
Detector
Held-out label
Window midpoint (s)

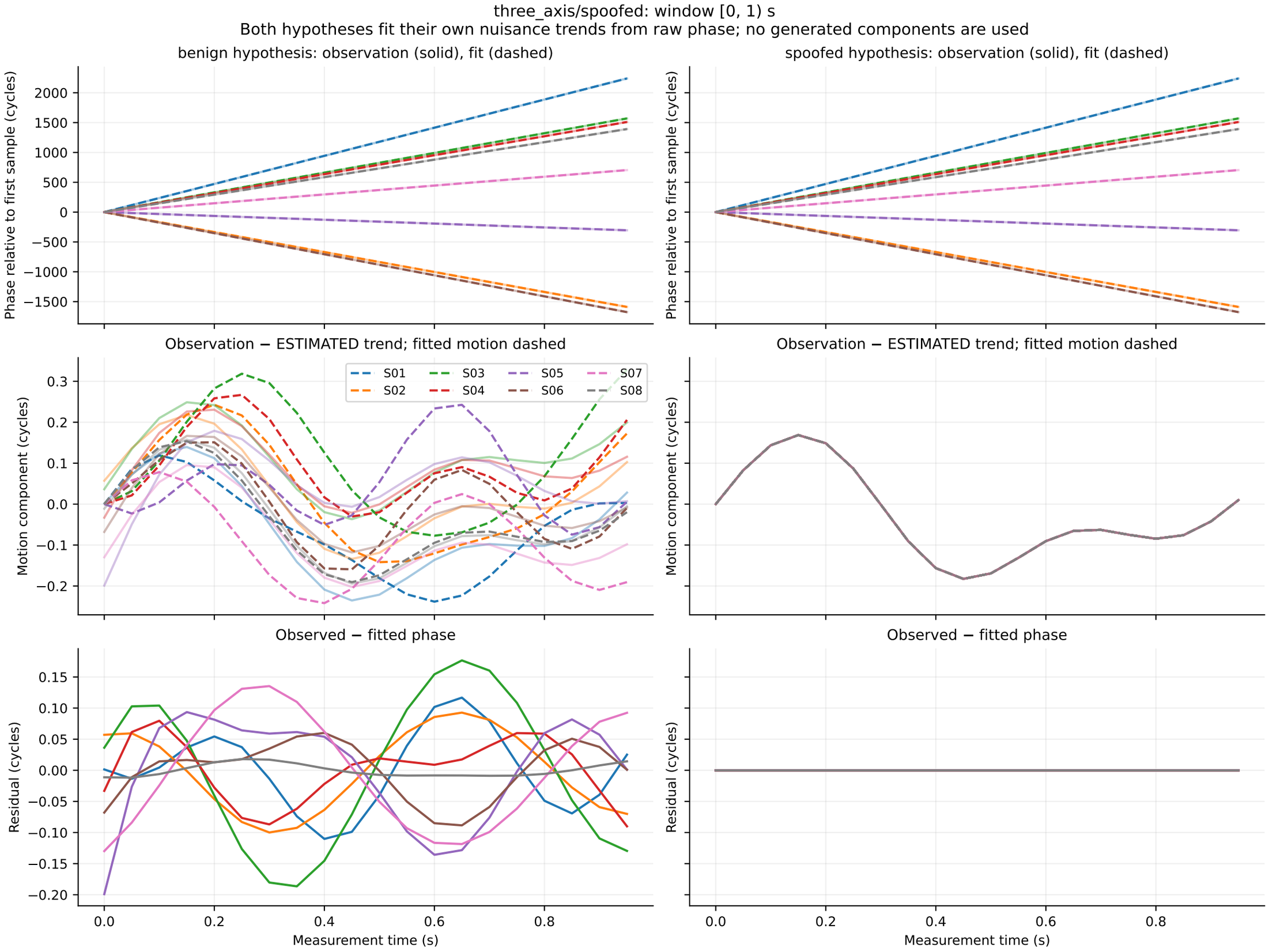

circle: oracle-displacement detector
Inconclusive results are retained; labels were read after prediction
benign: informative QR costs
spoofed: informative QR costs
Reduced negative log-likelihood
benign
spoofed
No reliable 3D fit
rank_deficient_motion
Positive favors benign; negative favors spoofing
Gamma = J_sp − J_benign
Benign
Inconclusive
Spoofed
Detector
Held-out label
Window midpoint (s)

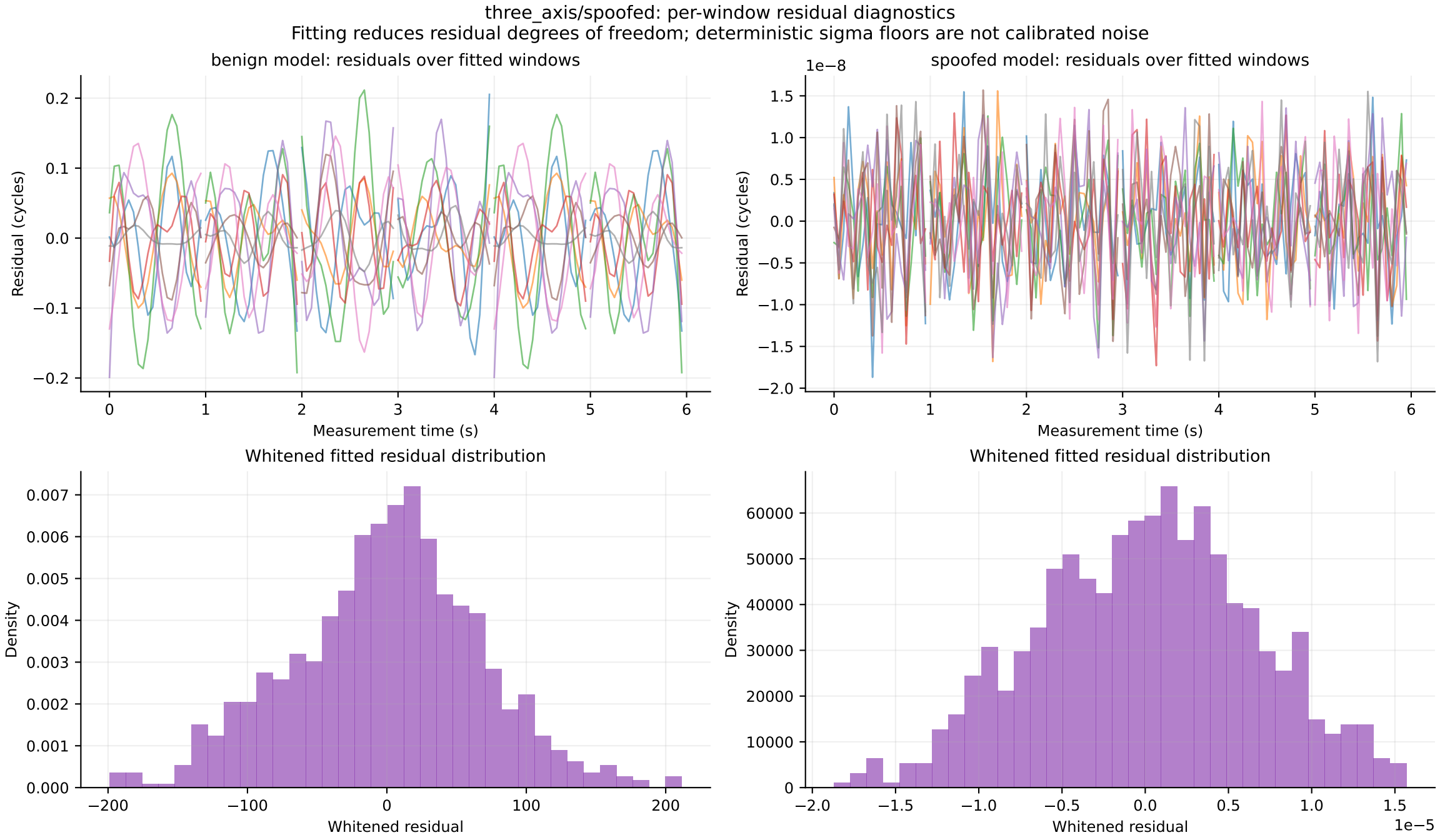

circle_coning: oracle-displacement detector
Inconclusive results are retained; labels were read after prediction
benign: informative QR costs
spoofed: informative QR costs
Reduced negative log-likelihood
benign
spoofed
No reliable 3D fit
rank_deficient_motion
Positive favors benign; negative favors spoofing
Gamma = J_sp − J_benign
Benign
Inconclusive
Spoofed
Detector
Held-out label
Window midpoint (s)

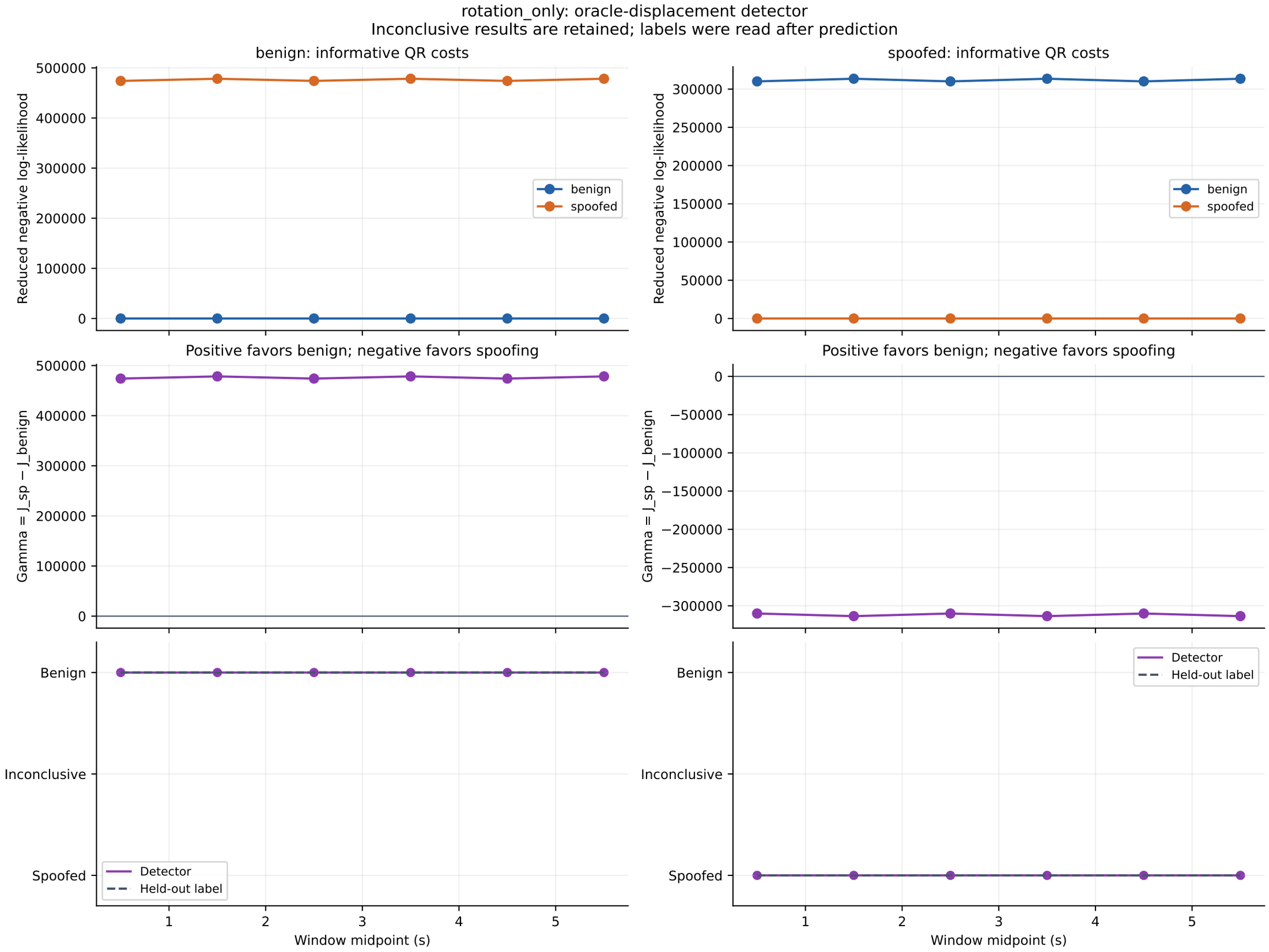

three_axis: oracle-displacement detector
Inconclusive results are retained; labels were read after prediction
benign: informative QR costs
spoofed: informative QR costs
Reduced negative log-likelihood
benign
spoofed
Positive favors benign; negative favors spoofing
Gamma = J_sp − J_benign
Benign
Inconclusive
Spoofed
Detector
Held-out label
Window midpoint (s)

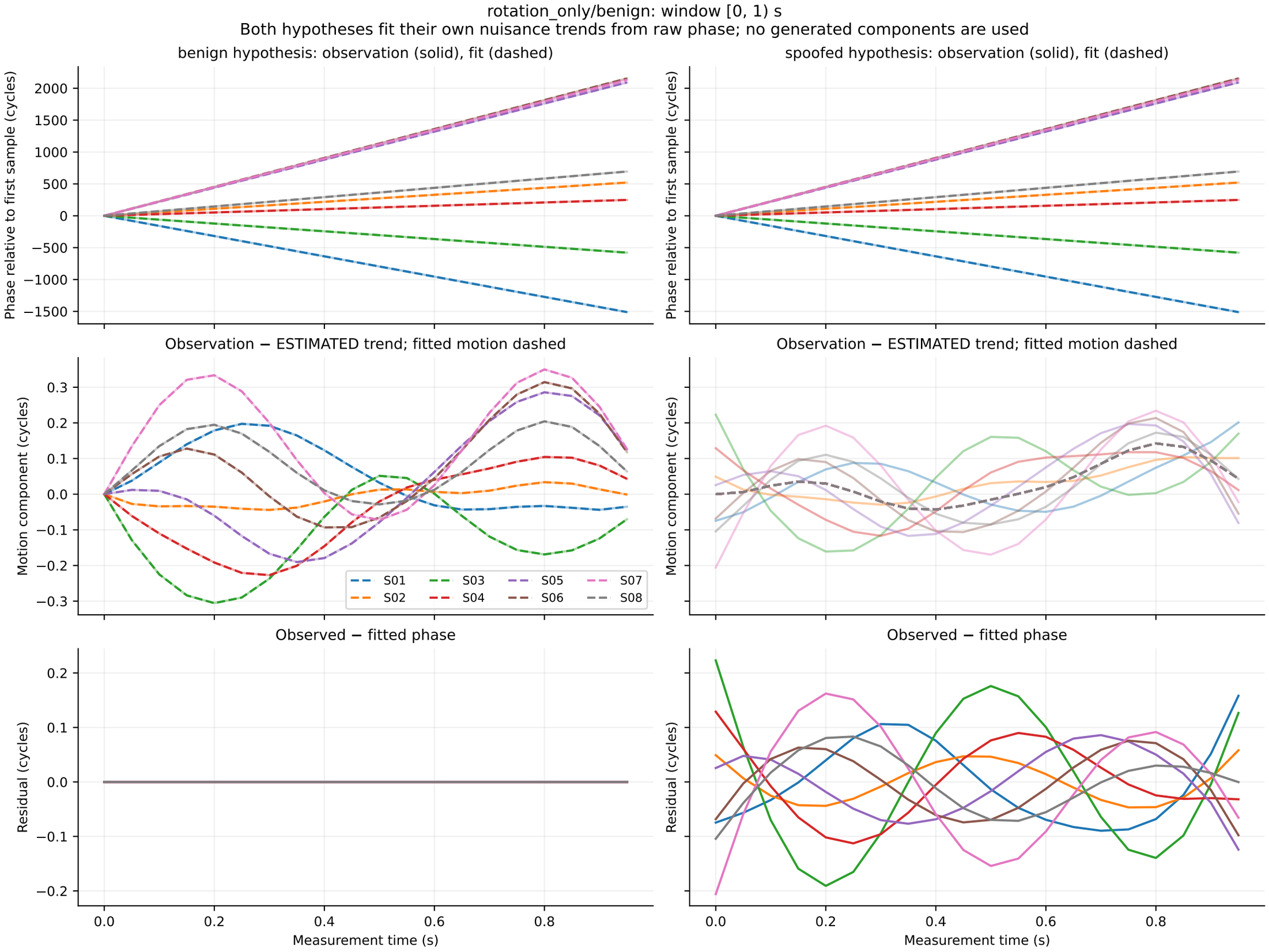

three_axis/benign: window [0, 1) s
Both hypotheses fit their own nuisance trends from raw phase; no generated components are used
benign hypothesis: observation (solid), fit (dashed)
spoofed hypothesis: observation (solid), fit (dashed)
Phase relative to first sample (cycles)
Observation − ESTIMATED trend; fitted motion dashed
Motion component (cycles)
S01
S02
S03
S04
S05
S06
S07
S08
Observed − fitted phase
Residual (cycles)
Measurement time (s)

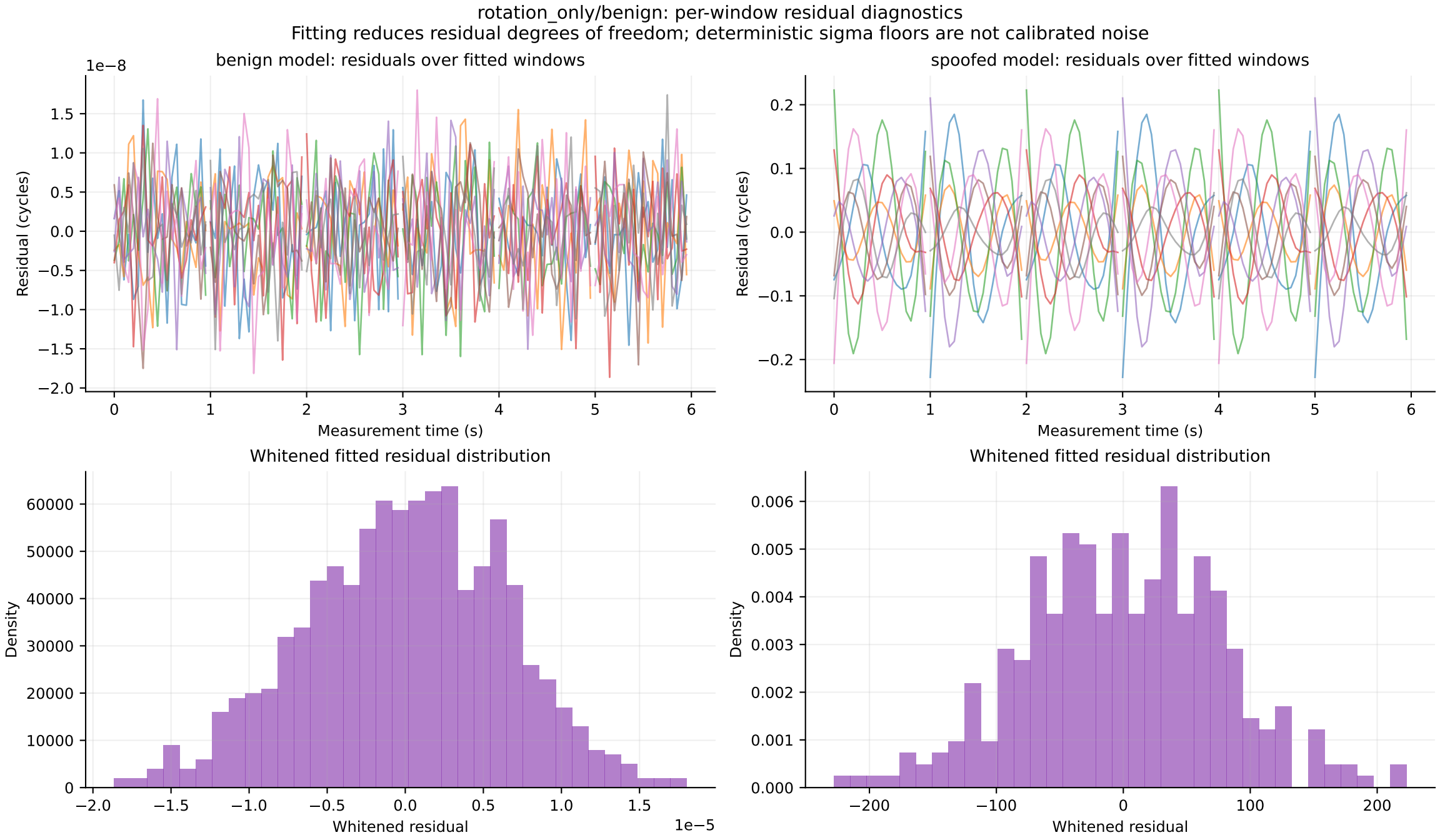

three_axis/benign: per-window residual diagnostics
Fitting reduces residual degrees of freedom; deterministic sigma floors are not calibrated noise
benign model: residuals over fitted windows
1e−8
Residual (cycles)
Measurement time (s)
spoofed model: residuals over fitted windows
Residual (cycles)
Measurement time (s)
Whitened fitted residual distribution
Density
Whitened residual
1e−5
Whitened fitted residual distribution
Density
Whitened residual

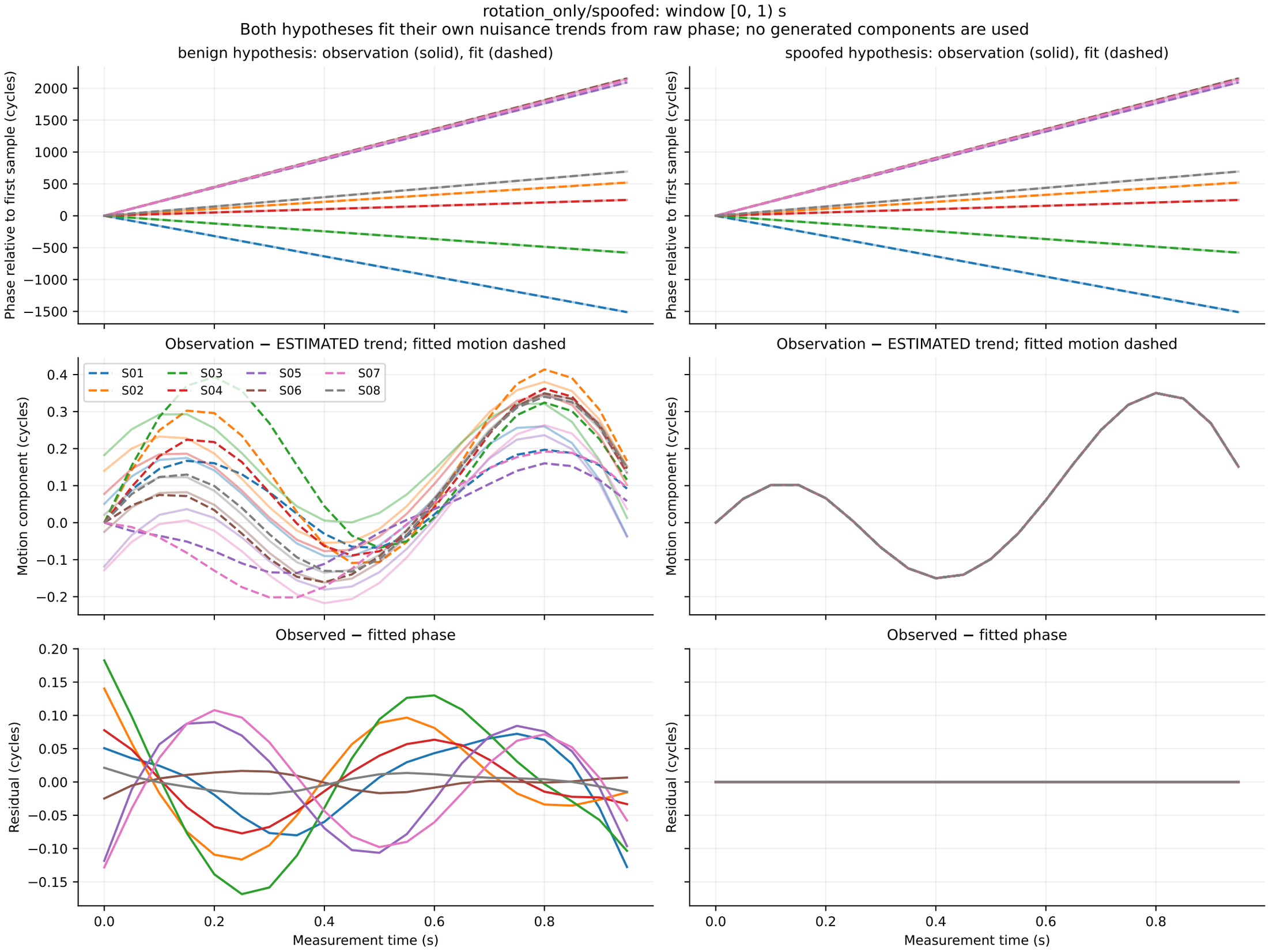

three_axis/spoofed: window [0, 1) s
Both hypotheses fit their own nuisance trends from raw phase; no generated components are used
benign hypothesis: observation (solid), fit (dashed)
spoofed hypothesis: observation (solid), fit (dashed)
Phase relative to first sample (cycles)
Observation − ESTIMATED trend; fitted motion dashed
Motion component (cycles)
S01
S02
S03
S04
S05
S06
S07
S08
Observed − fitted phase
Residual (cycles)
Measurement time (s)

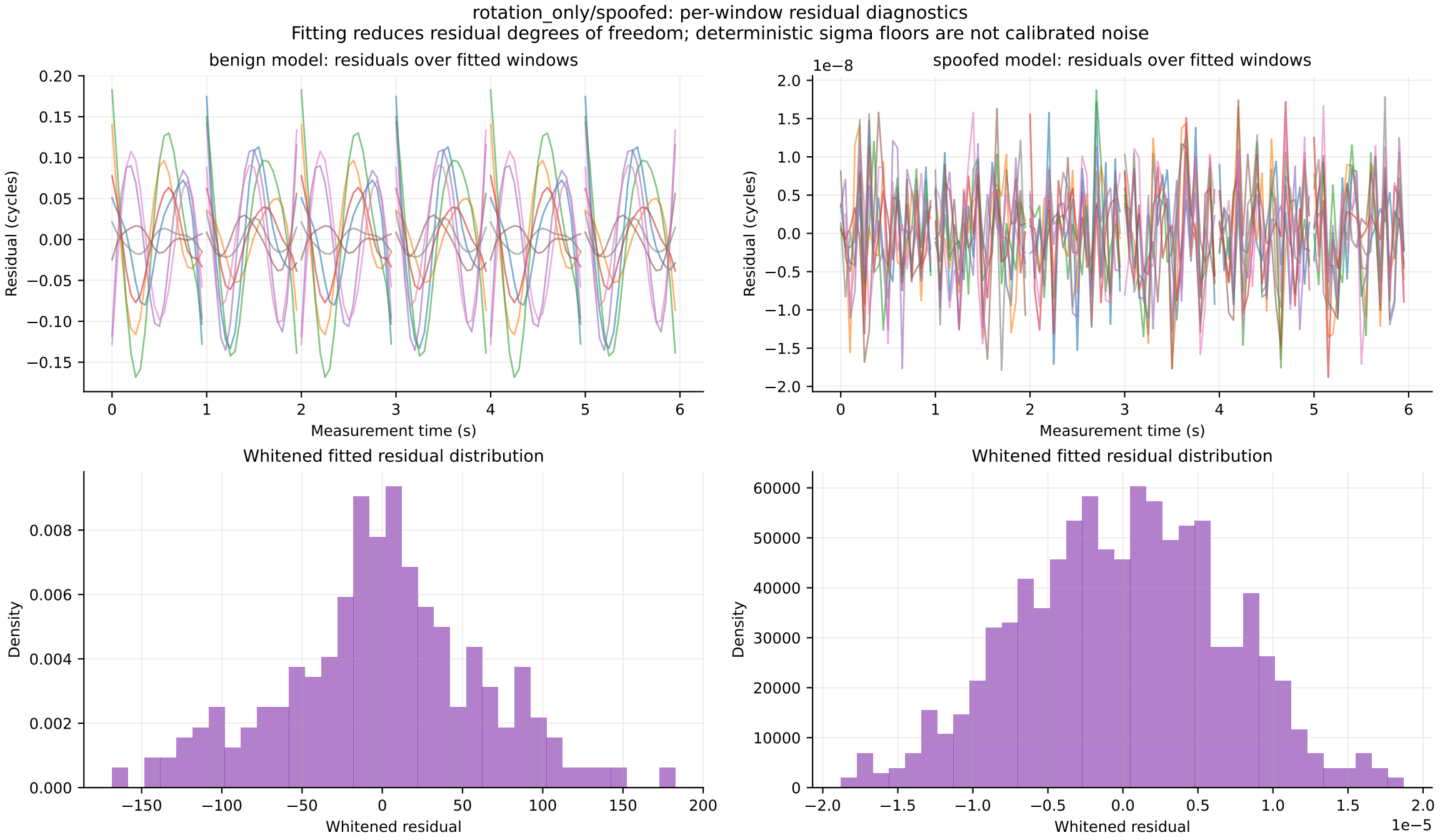

three_axis/spoofed: per-window residual diagnostics
Fitting reduces residual degrees of freedom; deterministic sigma floors are not calibrated noise
benign model: residuals over fitted windows
Residual (cycles)
Measurement time (s)
spoofed model: residuals over fitted windows
1e−8
Residual (cycles)
Measurement time (s)
Whitened fitted residual distribution
Density
Whitened residual
Whitened fitted residual distribution
Density
Whitened residual
1e−5

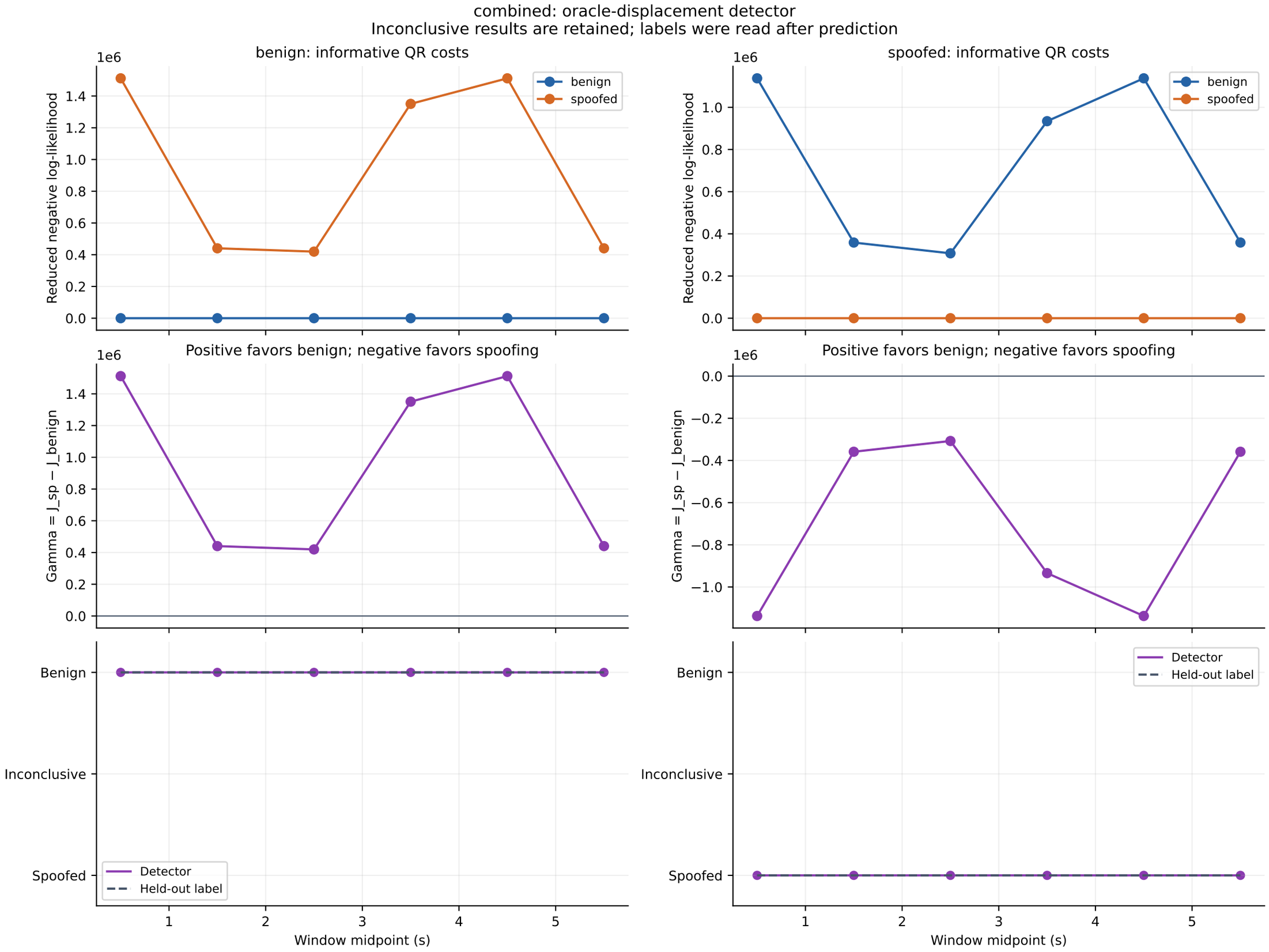

rotation_only: oracle-displacement detector
Inconclusive results are retained; labels were read after prediction
benign: informative QR costs
spoofed: informative QR costs
Reduced negative log-likelihood
benign
spoofed
Positive favors benign; negative favors spoofing
Gamma = J_sp − J_benign
Benign
Inconclusive
Spoofed
Detector
Held-out label
Window midpoint (s)

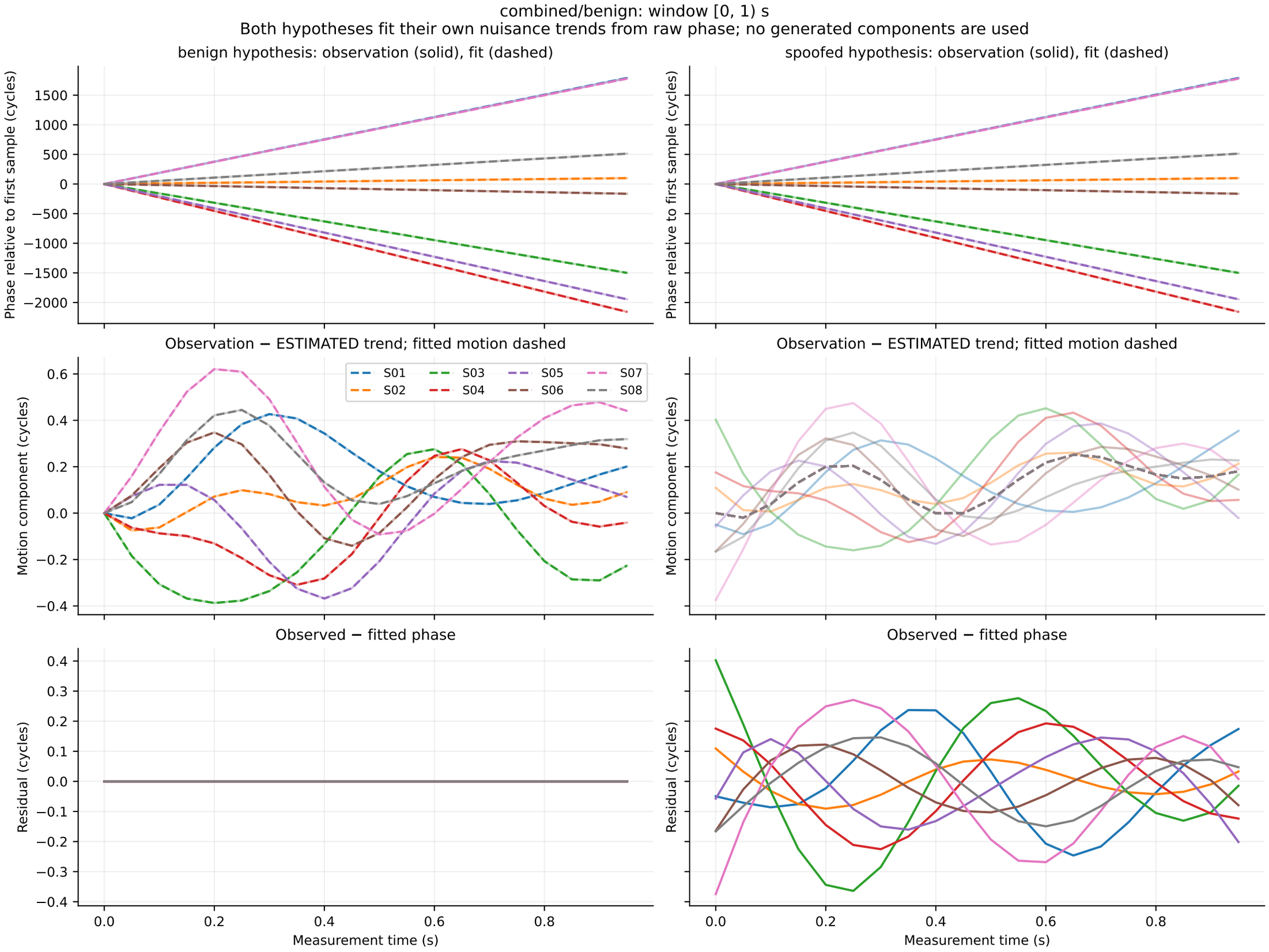

rotation_only/benign: window [0, 1) s
Both hypotheses fit their own nuisance trends from raw phase; no generated components are used
benign hypothesis: observation (solid), fit (dashed)
spoofed hypothesis: observation (solid), fit (dashed)
Phase relative to first sample (cycles)
Observation − ESTIMATED trend; fitted motion dashed
Motion component (cycles)
S01
S02
S03
S04
S05
S06
S07
S08
Observed − fitted phase
Residual (cycles)
Measurement time (s)

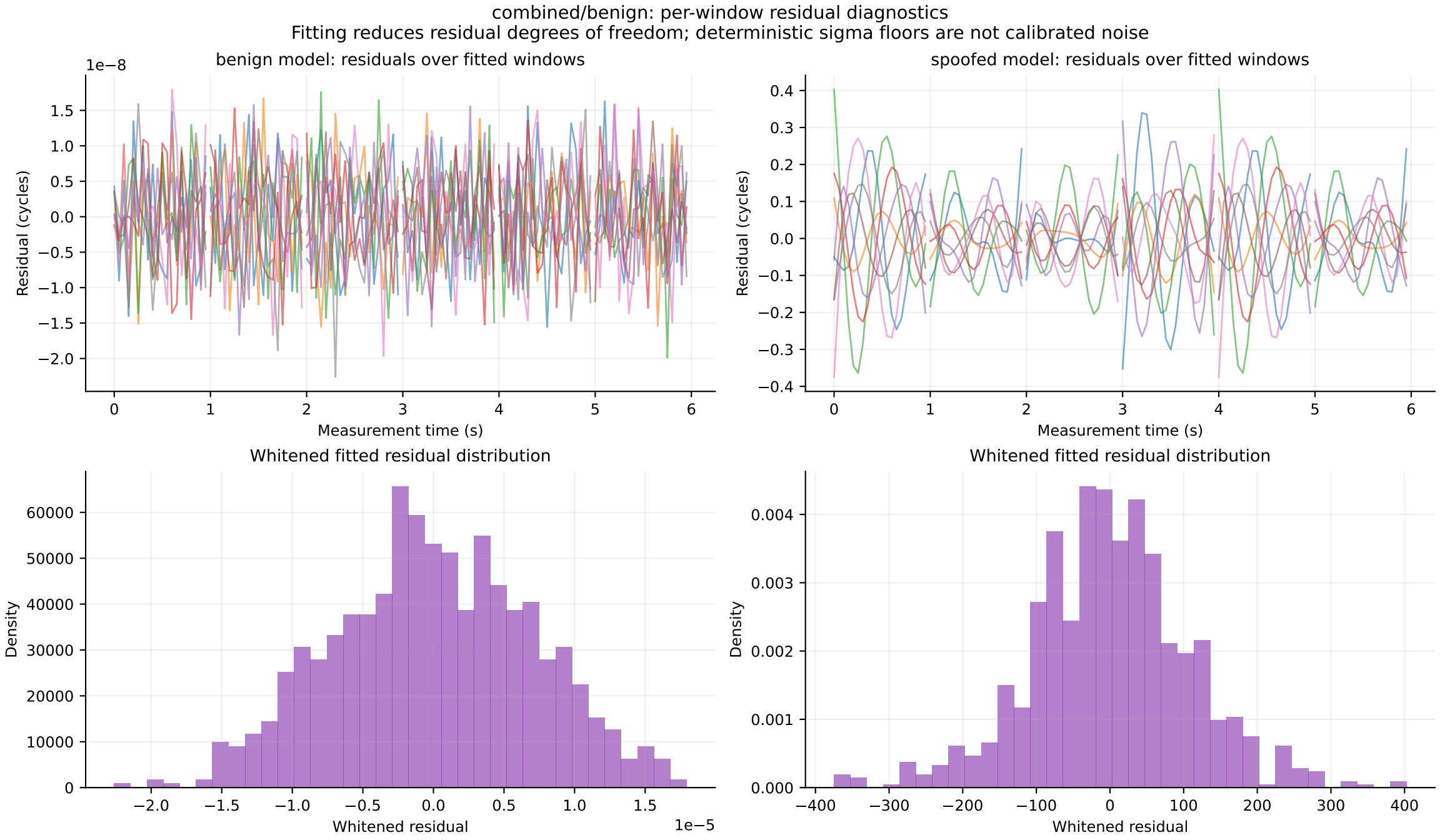

rotation_only/benign: per-window residual diagnostics
Fitting reduces residual degrees of freedom; deterministic sigma floors are not calibrated noise
benign model: residuals over fitted windows
1e−8
Residual (cycles)
Measurement time (s)
spoofed model: residuals over fitted windows
Residual (cycles)
Measurement time (s)
Whitened fitted residual distribution
Density
Whitened residual
1e−5
Whitened fitted residual distribution
Density
Whitened residual

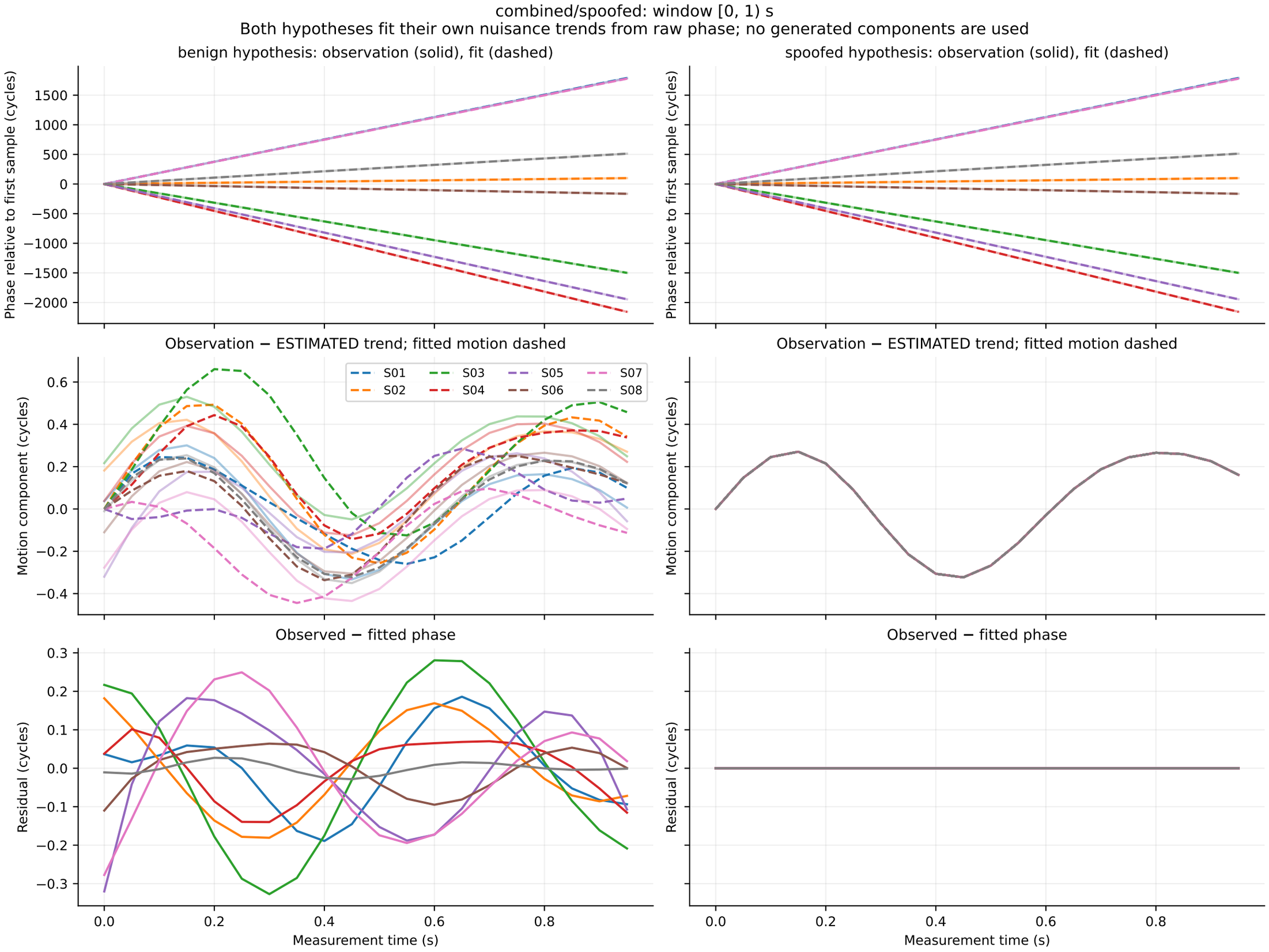

rotation_only/spoofed: window [0, 1) s
Both hypotheses fit their own nuisance trends from raw phase; no generated components are used
benign hypothesis: observation (solid), fit (dashed)
spoofed hypothesis: observation (solid), fit (dashed)
Phase relative to first sample (cycles)
Observation − ESTIMATED trend; fitted motion dashed
Motion component (cycles)
S01
S02
S03
S04
S05
S06
S07
S08
Observed − fitted phase
Residual (cycles)
Measurement time (s)

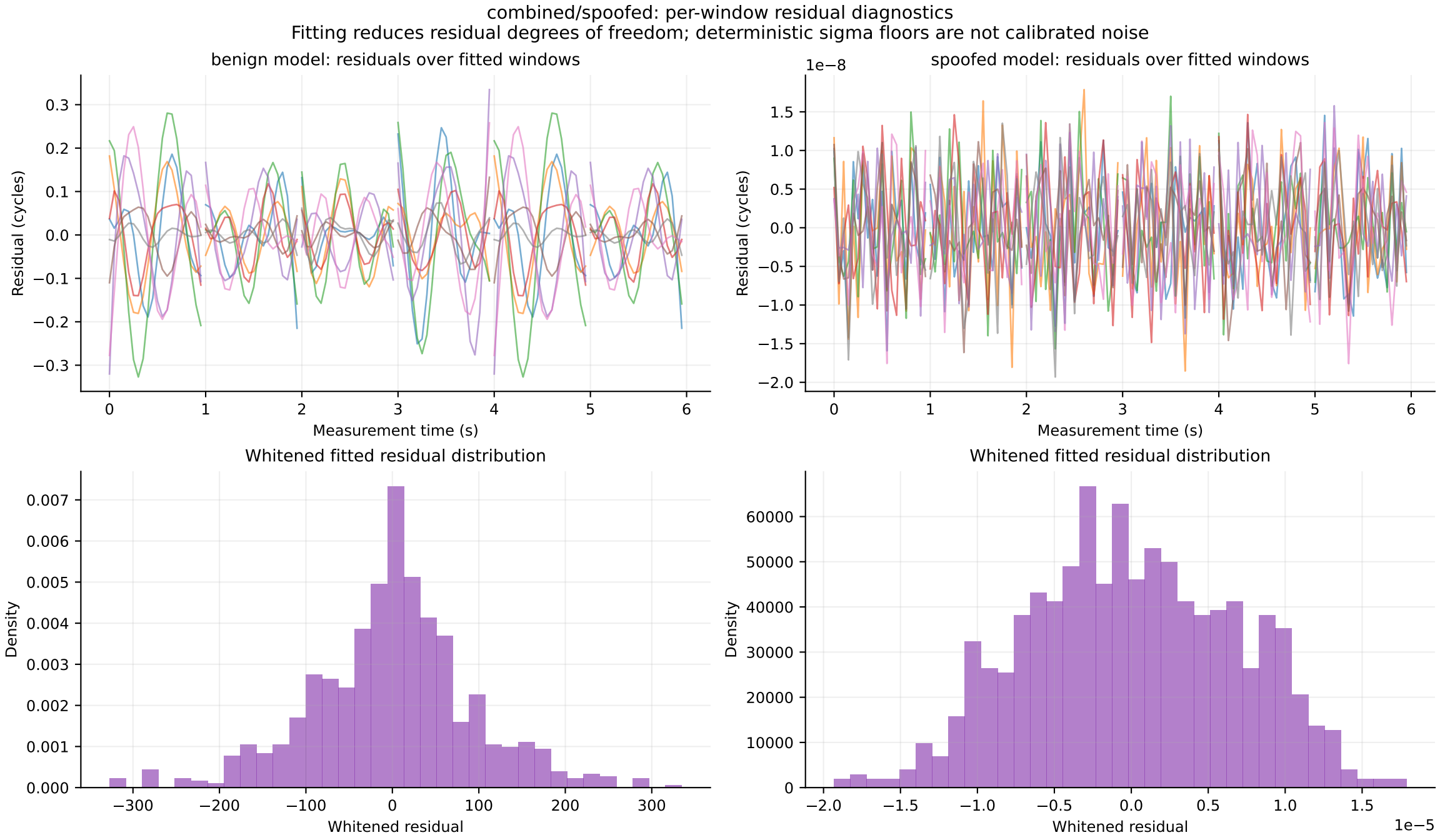

rotation_only/spoofed: per-window residual diagnostics
Fitting reduces residual degrees of freedom; deterministic sigma floors are not calibrated noise
benign model: residuals over fitted windows
Residual (cycles)
Measurement time (s)
spoofed model: residuals over fitted windows
1e−8
Residual (cycles)
Measurement time (s)
Whitened fitted residual distribution
Density
Whitened residual
Whitened fitted residual distribution
Density
Whitened residual
1e−5


combined: oracle-displacement detector
Inconclusive results are retained; labels were read after prediction
benign: informative QR costs
spoofed: informative QR costs
1e6
Reduced negative log-likelihood
benign
spoofed
Positive favors benign; negative favors spoofing
Gamma = J_sp − J_benign
Benign
Inconclusive
Spoofed
Detector
Held-out label
Window midpoint (s)


combined/benign: window [0, 1) s
Both hypotheses fit their own nuisance trends from raw phase; no generated components are used
benign hypothesis: observation (solid), fit (dashed)
spoofed hypothesis: observation (solid), fit (dashed)
Phase relative to first sample (cycles)
Observation − ESTIMATED trend; fitted motion dashed
Motion component (cycles)
S01
S02
S03
S04
S05
S06
S07
S08
Observed − fitted phase
Residual (cycles)
Measurement time (s)


combined/benign: per-window residual diagnostics
Fitting reduces residual degrees of freedom; deterministic sigma floors are not calibrated noise
benign model: residuals over fitted windows
1e−8
Residual (cycles)
Measurement time (s)
spoofed model: residuals over fitted windows
Residual (cycles)
Measurement time (s)
Whitened fitted residual distribution
Density
Whitened residual
1e−5
Whitened fitted residual distribution
Density
Whitened residual


combined/spoofed: window [0, 1) s
Both hypotheses fit their own nuisance trends from raw phase; no generated components are used
benign hypothesis: observation (solid), fit (dashed)
spoofed hypothesis: observation (solid), fit (dashed)
Phase relative to first sample (cycles)
Observation − ESTIMATED trend; fitted motion dashed
Motion component (cycles)
S01
S02
S03
S04
S05
S06
S07
S08
Observed − fitted phase
Residual (cycles)
Measurement time (s)


combined/spoofed: per-window residual diagnostics
Fitting reduces residual degrees of freedom; deterministic sigma floors are not calibrated noise
benign model: residuals over fitted windows
Residual (cycles)
Measurement time (s)
spoofed model: residuals over fitted windows
1e−8
Residual (cycles)
Measurement time (s)
Whitened fitted residual distribution
Density
Whitened residual
Whitened fitted residual distribution
Density
Whitened residual
1e−5